Predicting Hospital Readmission for Patients with Diabetes

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Agenda

- Problem Statement
- Class Imbalance & Modeling Approach
- Hyperparameter Tuning
- Model Performance Evaluation
- Implementation Plan
- Additional Modeling Consideration

Problem Statement & Data Overview

The Cost of Hospital Readmissions for Patients with Diabetes

Problem Defined

Hospitals face significant **financial** and **clinical** challenges due to **30-day readmissions** among patients with **diabetes**

leading to

- Up to $217,000 annually in Medicare HRRP penalties per hospital
- Increased patient morbidity
- Reduced quality of care

Feature Assumption

- **Clinical History**
- **Demographics**
- **Discharge Information**
- **Prescribed Medicine**

These features potentially play an important role in **predicting patient readmission**

Goal

Developing models for predicting patients at risk for hospital readmission can enable:

- Early **identification of high-risk patients** with diabetes
- Interventions for **better long-term health outcomes and quality of care**
- **Reduced costs** for hospitals, insurance companies, and patients for readmitting to the hospital

Data Overview – Leveraging Data to Build Predictive Models

Data Source: Ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks.

Features

Clinical Factors	Hospital stay duration, lab tests, medications, diagnoses
Demographics	Age, race, gender
Admission & Discharge	Type, source, disposition
Medication History	Insulin, metformin, other diabetes drugs
Target Variable	**Readmission (<30 days)**

General Observations

Data Quality: Missing values, high dimensionality

Complexity: Multiple comorbidities & external factors influencing readmission

Class Imbalance: Roughly 11% membership for minority class

Modeling Considerations: Balancing model accuracy and interpretability for clinical setting

Reducing Categorical Features' Dimensionality

Many categorical features had **high cardinality**, leading to increased complexity and potential overfitting.

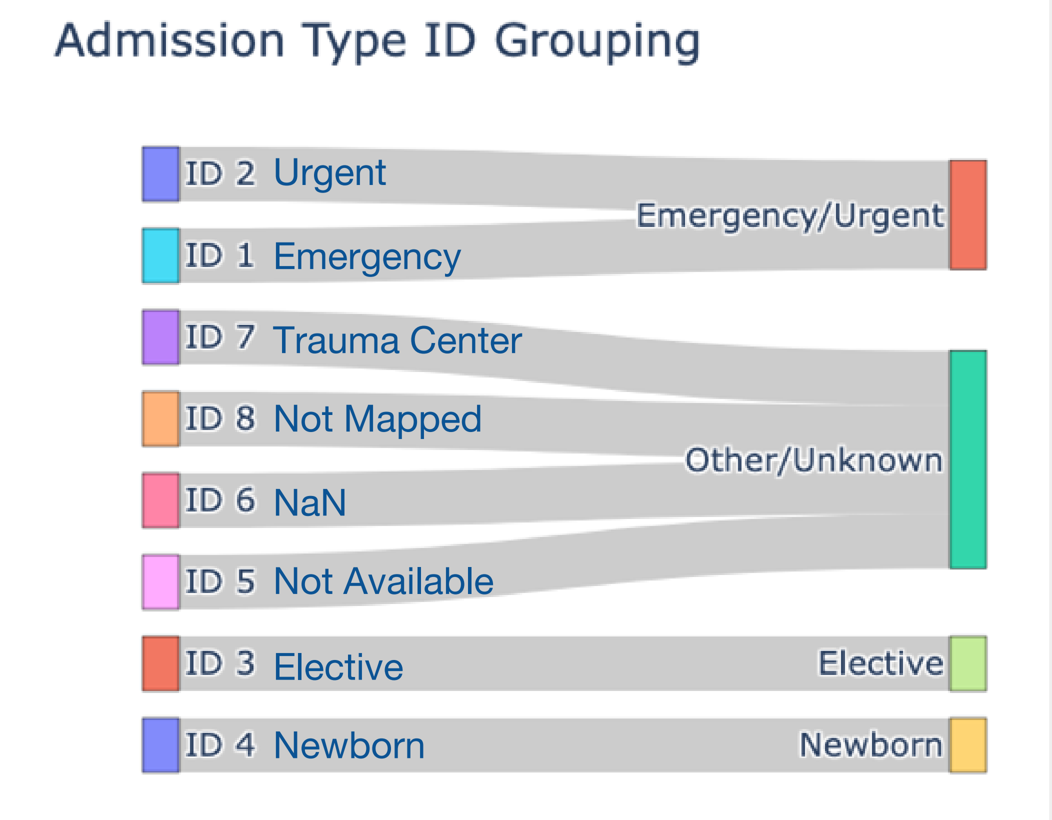
		Reduced to	
admission_type_id	8	4	Contained **numerous detailed** categories that were **not significantly distinct** for prediction
admission_source_id	25	5	
discharge_disposition_id	30	5	

Method: To reduce dimensionality while preserving meaningful distinctions, categorical variables were grouped by **merging similar categories** based on clinical and predictive relevance

diag_1	##	9	Hundreds of **unique ICD-9 codes**, making them **difficult** to interpret and use in modeling.
diag_2	##	9	
diag_3	##	9	

Method: Instead of handling hundreds of unique ICD-9 codes, diagnoses were grouped into **9 disease categories** based on **clinical significance** (See Appendix)

Example Grouping: admission_type_id



Engineering Features for Risk Signaling for Readmission

To enhance risk signaling for hospital readmission, we engineered additional features by leveraging domain knowledge.

Medication Management

on_multiple_meds	Binary indicator for patients on 3+ medications
med_changes_total	Count of medication changes (increases or decreases)
insulin_binary	Flag for patients using insulin

Patient Risk Profile

age_risk	Age risk
risk_score Composite risk score by this formula: ***risk_score* = *w_A+(w_I×2)+(w_E×1.5)+(w_M×1.2)***	Age risk
	Inpatient history (weighted x2)
	Emergency visits (weighted x1.5)
	Medication count (weighted x1.2)

Hospital Utilization

total_visits	Sum of outpatient, emergency, and inpatient visits
repeated_emergency	Flag for patients with multiple emergency visits
all_usage	Flag for patients who utilized all three care settings

Admission Characteristics

is_emergency	Binary indicator for urgent/emergency admissions

Class Imbalance & Modeling Approach

Class Imbalance in Readmissions – A Challenge for Prediction

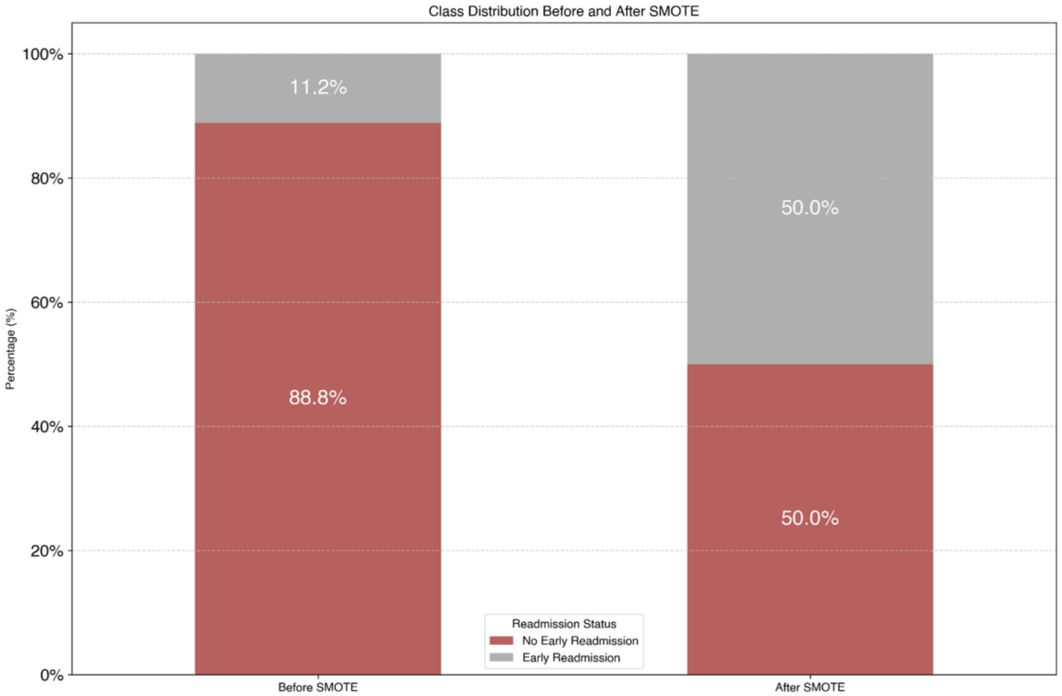
Standard models **assume balanced data**, causing poor predictive performance on the minority class (readmissions)

- A model could achieve high accuracy by defaulting to the majority class (non-readmissions), missing high-risk patients

SMOTE (Synthetic Minority Oversampling Technique)	Generates synthetic samples for the minority class → **can lead to overfitting**
Class Weights	Helps models tune performance in predicting the minority class → **sensitive to outliers**

Using SMOTE and class weights together can sometimes degrade performance due to compounding biases and overfitting

Class Imbalance Before v.s. After SMOTE



Evaluating SMOTENC & Class Weights Across Models

We will test both **SMOTENC** and **Class Weights** across multiple models to assess their effectiveness in handling class imbalance.

Model	Purpose	Strengths
Logistic Regression	Baseline model	Simple, interpretable
Random Forest	Uses bagging to reduce variance	Effective for class imbalance, improves generalization
XGBoost	Corrects errors iteratively	Captures complex relationships, improves minority class performance
LightGBM	Gradient boosting optimized for speed	Handles large datasets efficiently, reduces overfitting
CatBoost	Boosting method optimized for categorical features	Works well with categorical data, less hyperparameter tuning needed

Hyperparameter Tuning

Framework for Developing Models for Cost-Optimal Readmission Prevention

Hyperparameter tuning per model iteration → **Model calibration to real world probabilities** → **Threshold tuning in accordance with financial assumptions***

Baseline Models	Hyperparameter Tuning	Calibration	Threshold Tuning	Best Performing Models
Logistic Regression **Random Forest** **XGBoost** **LightGBM** **CatBoost**	Ensuring each model is optimized **independently** for **predictive performance (F2)**	Adjusting **probability outputs** so they align with **actual readmission rates**	Selecting decision thresholds to **optimize cost effectiveness** in <u>readmission reduction</u>	• **F2 Optimization** • **Best Financial Performance** **XGBoost (SMOTE)** **XGBoost (Class Weights)** **CatBoost (Class Weights)**

** Tuning hyperparameters in accordance with financial assumptions is computationally expensive and would require re-tuning whenever costs change. We agreed on starting with finding the hyperparameters that achieve the best F2 score*

Tuning our Models for Predicting Readmission

Hyperparameter Tuning	Maximize F2 Score (favoring recall to prioritize flagging at-risk patients)

Metric Selection Rationale - Choosing the Right Metric for Imbalanced Data
- Recall: Measures how many actual positives were correctly identified
- F2 score : Prioritizes recall over precision

$$F2 = \frac{5 \times \text{Precision} \times \text{Recall}}{4 \times \text{Precision} + \text{Recall}}$$

Parameters

1. **Learning rate**
   Controls how quickly models adapt to errors
2. **Tree depth**
   controls complexity of individual trees
3. **Regularization parameters**
   prevents overfitting

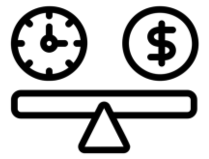
Outcome

Optimized for identifying at-risk patients but doesn't align with financial outcomes

Why Probability Calibration Matters in Clinical Settings

Calibration	A correction technique that aligns predicted probabilities with actual observed outcomes

Why is calibration needed?



For hospital readmission models, improperly calibrated probabilities from models could lead to **over-** or **under-predicting** readmission risk, impacting patient care and financial costs.

How is calibration done?

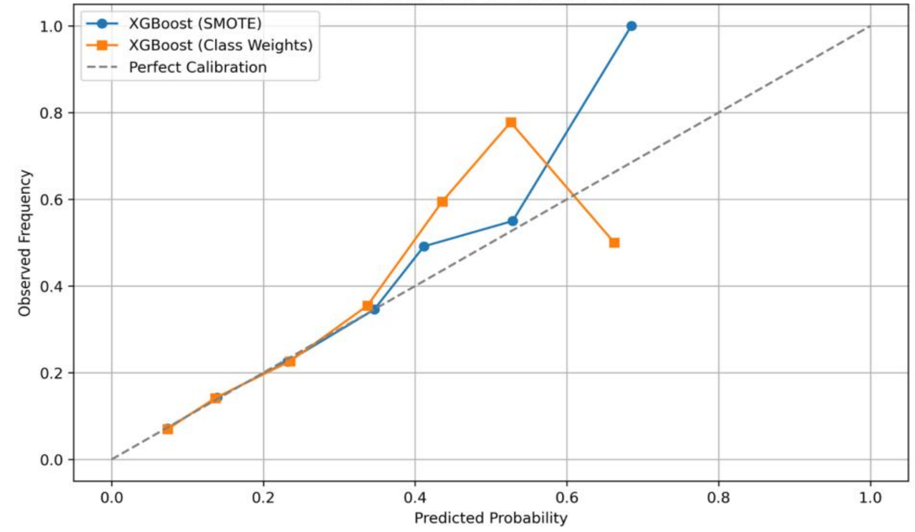
Isotonic Regression	Non-parametric method that finds a non-decreasing function to map predicted probabilities to calibrated ones via: ● CalibratedClassifierCV or ● Custom wrapper (for CatBoost)

Takeaway: These models work well for general predictions, but clinical decision-making requires precise probability calibration to avoid over/underestimating risk.

XGBoost Calibration Analysis

As seen in the XGBoost calibration curve, both models overestimate probabilities at high confidence levels, leading to overconfident predictions that may mislead decision-making.

Calibration Curve for XGBoost Models



Evaluating Model Performance: The Precision-Recall Tradeoff

Insights

Class Weights outperform SMOTE on average

- 4 out of 5 models (Random Forest, CatBoost, Logistic Regression, and LightGBM)

RandomForest delivers the highest F2 scores

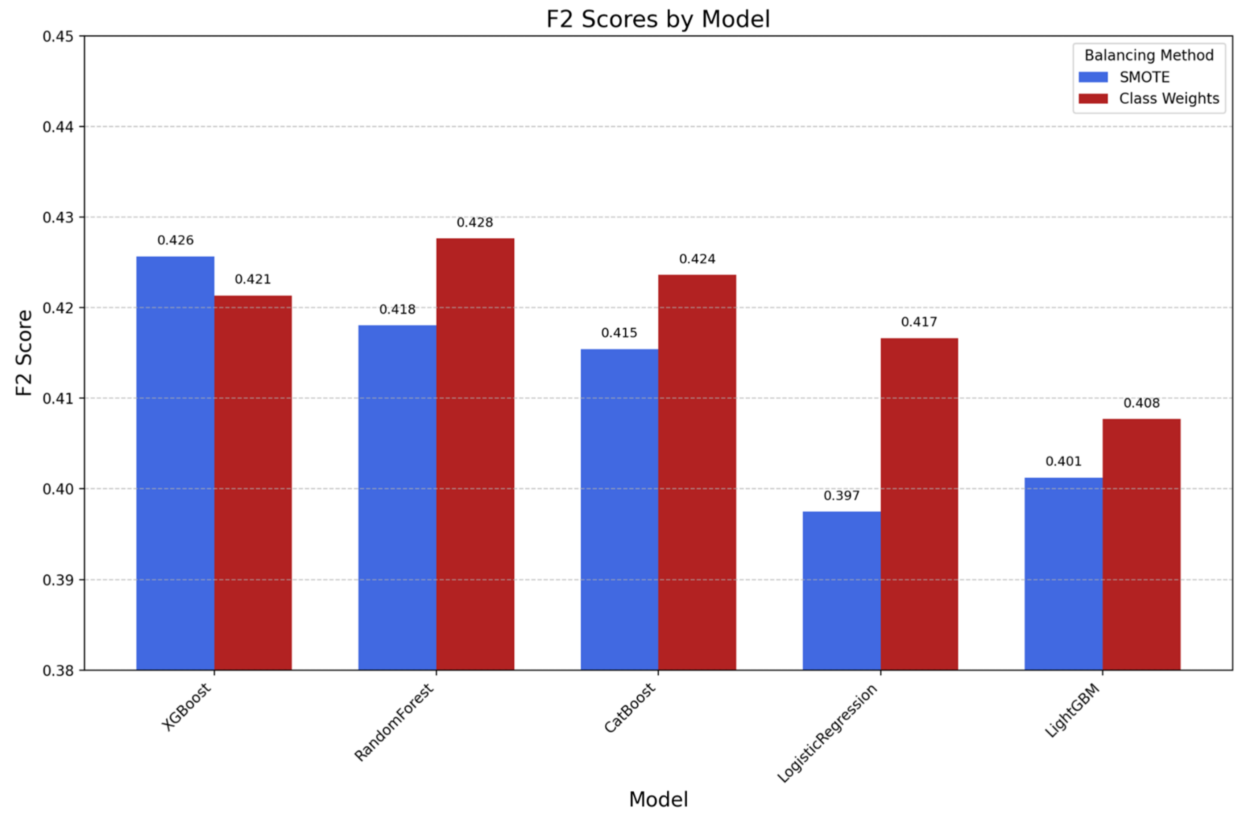
- Excels with class weighting, hurt by SMOTE

Weighing Model Performance Against Clinical Application:

Optimizing F2 score led to probability thresholds classifying patients at risk for readmission between:

Probability Threshold	0.08	0.10

F2 Scores by Model

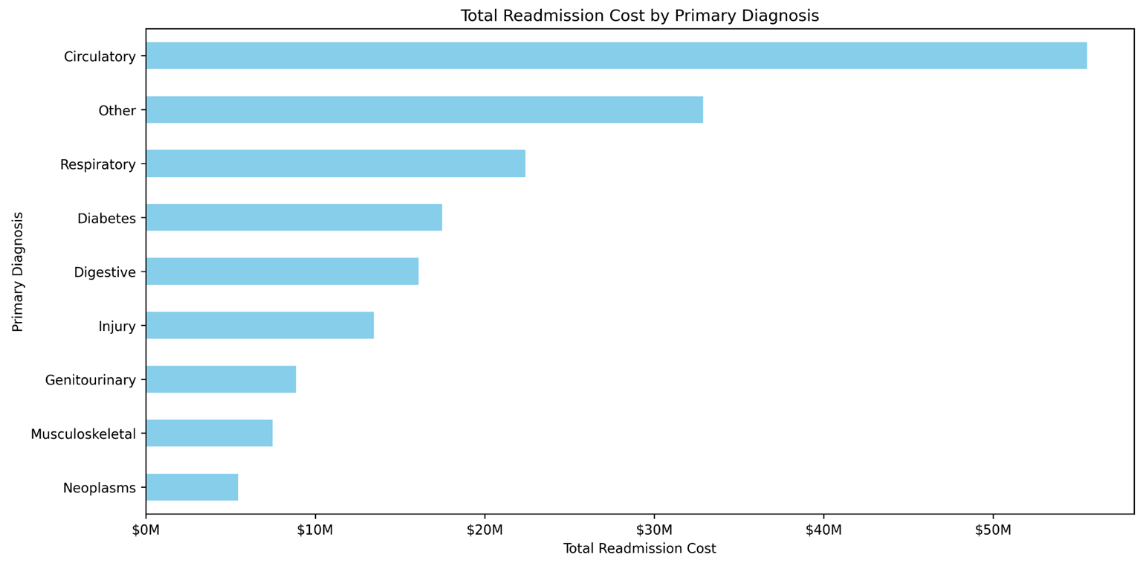


Incorporating Cost-Benefit Optimization – Threshold Tuning

Financial assumptions

Avg. Each Readmission Costs the Healthcare System	$16,300
Translates to $180M spent across the scope of this project	
Per Patient Intervention Costs the Healthcare System	$2,500
The equivalent an extra night spent in the hospital	
Intervention effectiveness (IE)	[30%, 40%, 50%] reduction in readmission risk

Total Readmission Cost by Primary Diagnosis



Healthcare savings

By preventing IE(%) of true readmissions ($16,300 each) while spending $2,500 on each high-risk patient (both correct and false alarms).

Savings = (TP × IE × $16,300) - ((TP + FP) × $2,500)

Evaluating Model Performance: Financial Impact

Threshold Testing	Tests threshold for each model **assigning the threshold that maximizes healthcare costs savings**

Consistent Performer (Preferred Model)

XGBoost (Class Weights)
Offers consistent savings at all intervention levels

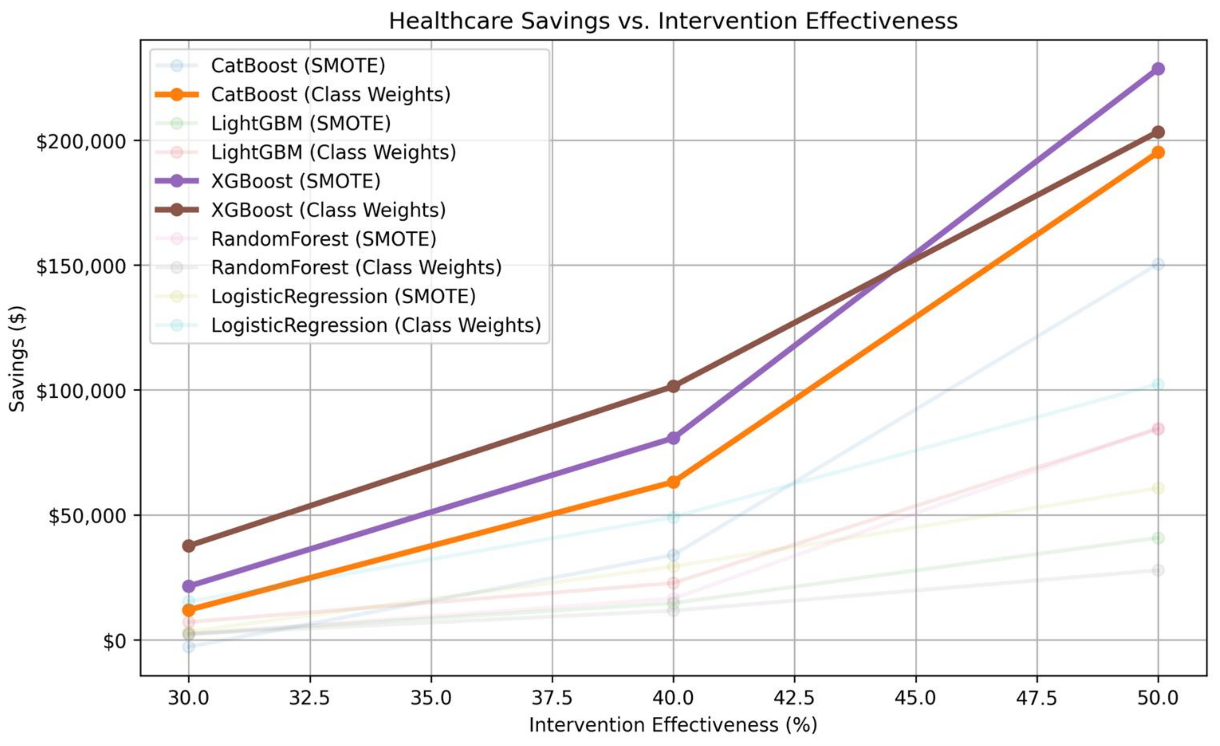
Delivering the Highest Savings ($)

XGBoost (SMOTE)	
Reaching $200,000 Savings	@ 50% Intervention Effectiveness

Takeaways: SMOTE Models underperform on average – Class weighting works better than oversampling for financial gains at modest intervention effectiveness levels
Higher intervention effectiveness → savings

- Savings increase as effectiveness rises from 30% to 50%

Healthcare Savings vs. Intervention Effectiveness



Optimizing Decision Thresholds for Preventing Hospital Readmissions Using XGBoost

XGBoost – increases financial performance while lowering the threshold for intervention support as effectiveness increases

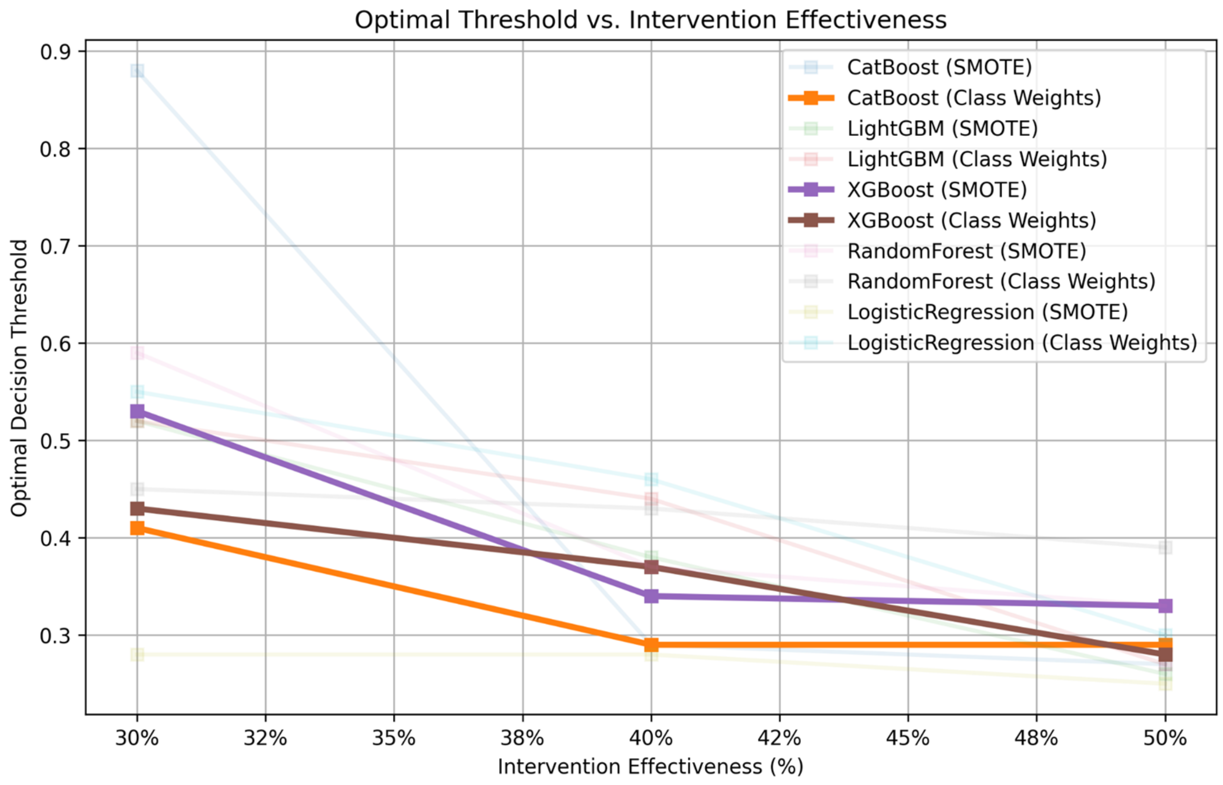
Takeaway: Most models take a conservative approach to preventing readmissions at first

- At 30% effectiveness, many models start with high decision thresholds (~70%), meaning they only intervene in high-risk cases to prevent unnecessary costs.
- Models lower their decision thresholds as intervention effectiveness rises from 30% to 50%, allowing more interventions

Preferred Model

XGBoost (Class Weights)	Effectively adjusts decision thresholds in response to increasing intervention effectiveness • Ensuring an optimal balance between cost savings and patient interventions • XGBoost (Class Weights) strikes a better balance—lowering thresholds sooner while maintaining financial efficiency.

Optimal Threshold vs. Intervention Effectiveness



Feature Importance – Risk Score, Key Indicator of Readmission

SHAP(Shapley Additive Explanations)

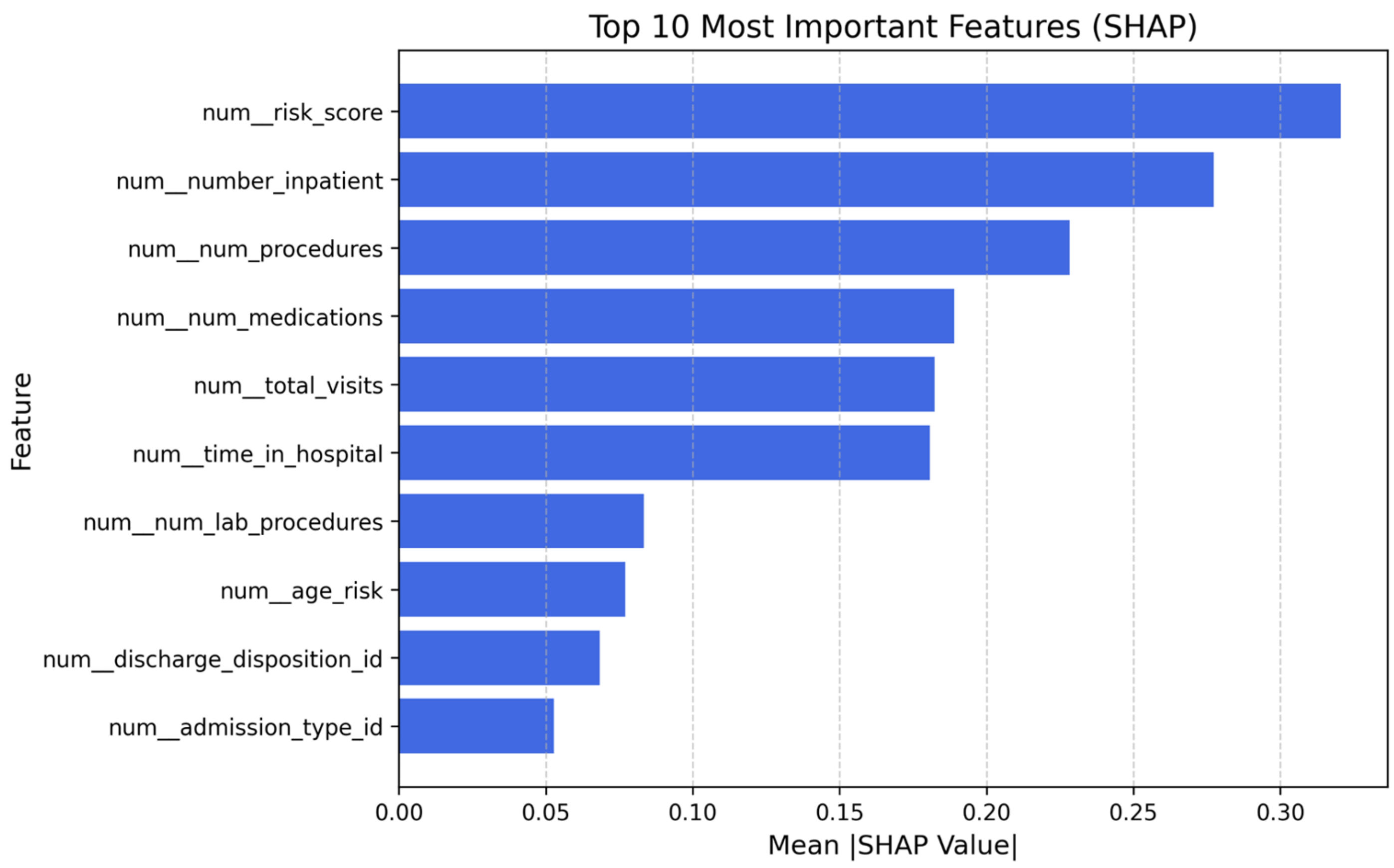
SHAP values measure how individual features impact the model's prediction.

- Higher SHAP values indicate a stronger influence on hospital readmission risk

Takeaways

- **Risk score** and **inpatient visit history** contribute most to readmission prediction
- **Lab procedures** and **discharge disposition** may provide additional predictive power.

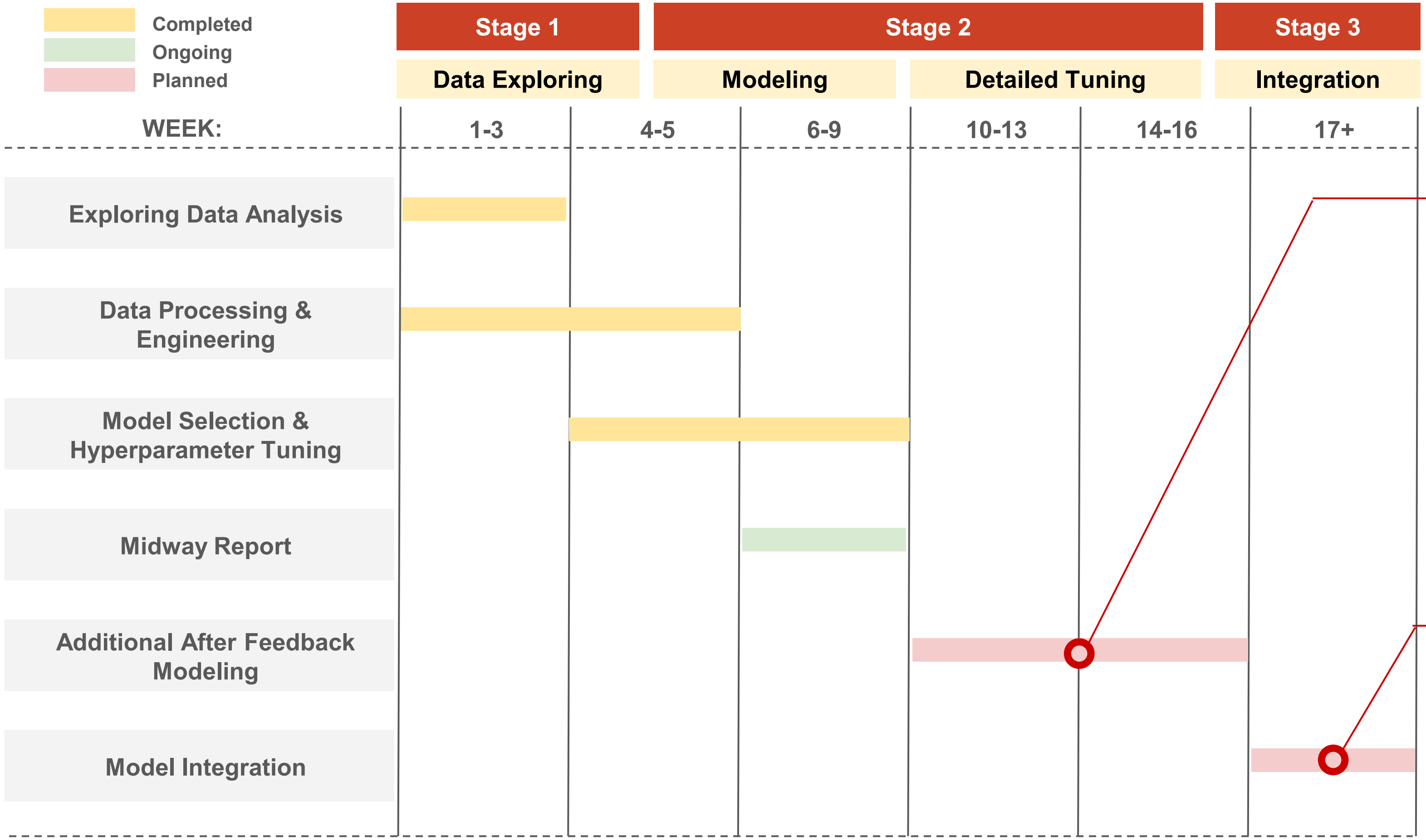
Top 10 Most Important Features (SHAP)



Implementation Plan

Implementation Plan: Integrating Predictions for Readmission Prevention

Implementation Timeline



Remaining Modeling Plan

1. **Further model calibration**
   a. Accurate risk probabilities
2. **Threshold tuning for region-specific financial constraints**
3. **Identify disparities in model performance**
   a. demographic
   b. patient groups

Implementation Plan

1. **Data Pipeline & System Integration**
   a. Automate data ingestion from hospital systems (e.g. Epic, Cerner)
2. **Model Deployment**
   a. Develop and implement a real-time prediction API to assess patient risk near discharge, including explainable outputs from the production model
3. **Clinical Workflow Integration**
   a. Embed predictions into discharge planning systems, working as an aid for healthcare professionals to evaluate discharge plans
4. **Monitoring & Continuous Improvement**
   a. Monitor model drift and performance and evaluate against cost-benefit framework

Additional Modeling Considerations

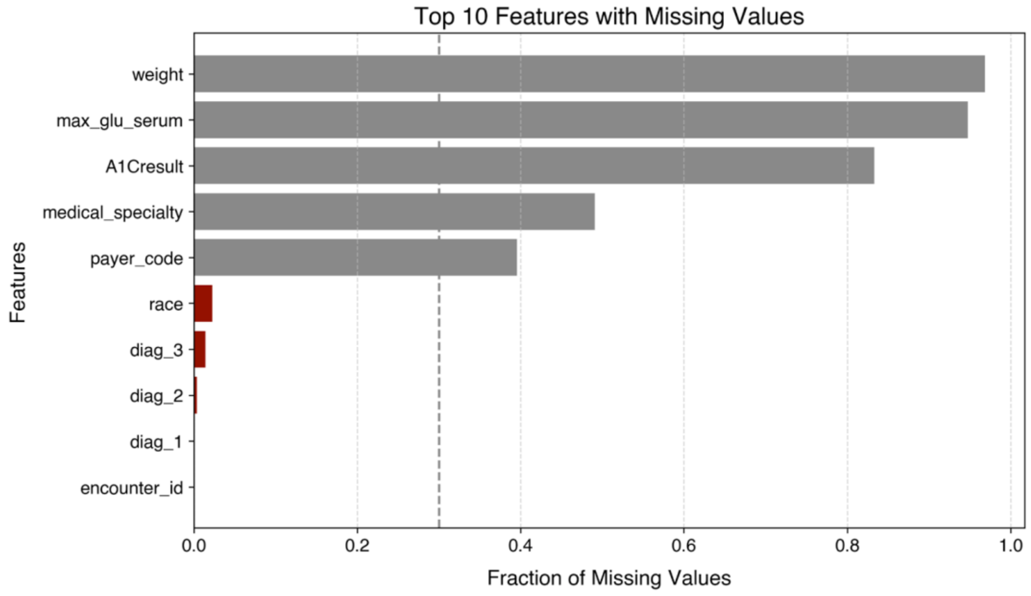
Limitations	Recommendations
Black-box models hinder interpretability in high-dimensional data.	Use advanced visualization techniques (e.g., parallel coordinates, Andrews curves).
Inconsistent feature importance across algorithms creates ambiguity.	Develop complementary rule-based models alongside black-box methods.
Using global class weights may ignore intra-group imbalance (e.g., across age groups, gender).	Apply feature-aware class weighting based on important categorical variables.
Naive imputation may not capture domain-specific nuances.	Apply Bayesian imputation with domain-inspired priors.
Raw probability outputs may be misinterpreted due to over-/under-confidence issues.	Integrate recalibration (e.g., isotonic regression) and explanation tools (e.g., SHAP).

Appendix

Data Overview – Quality & Target Variable

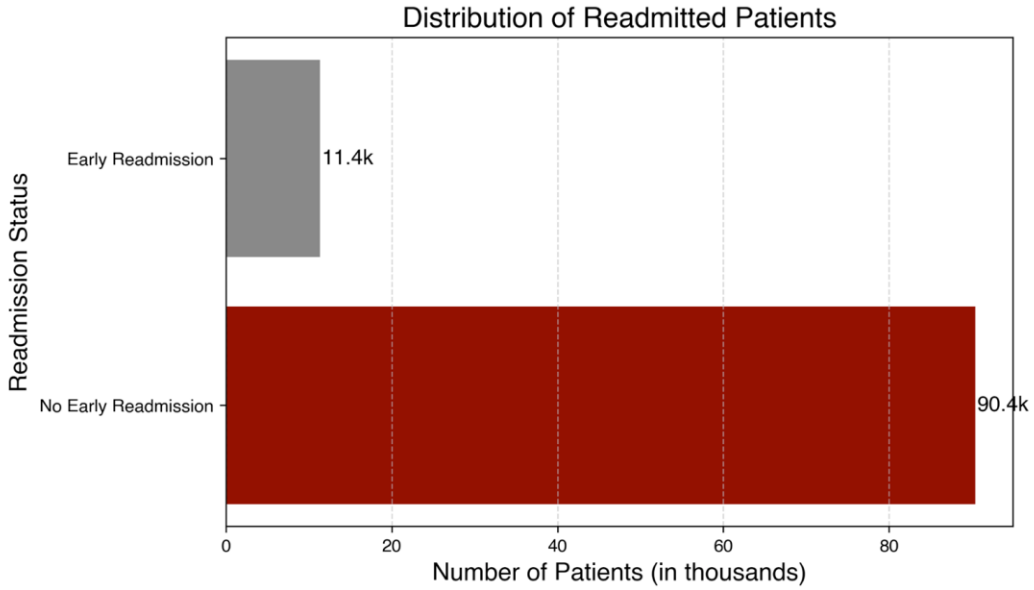
Addressing Data Quality

- 5 features with significant missing data
- Dropped rows with <3% missing data



Target Variable Early Hospital Readmission

- ~11% of patients with diabetes readmitted within 30 days, suggesting class imbalance



Alternative Method: LLM Driven Readmission Predictions

Why experiment with LLM driven approach?

- Traditional machine learning models suffer from class imbalance

Using llama3, readmission predictions can be made

- Without feature selection, hyperparameter tuning

However, coming with potential drawbacks:

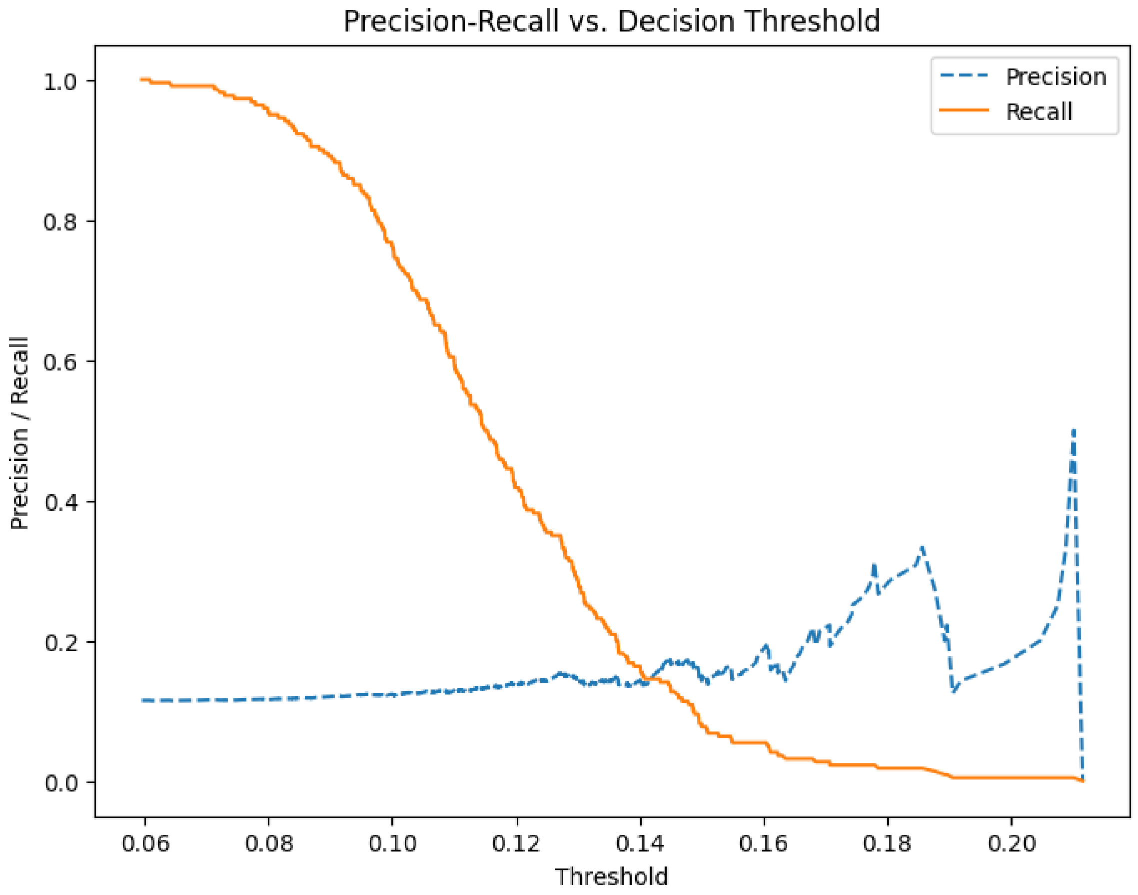
- Higher computational requirements than traditional models
- Potential data privacy concerns
- Less interpretability
- Lacks direct probability calibration

Performance in predicting readmission patients		
	Precision	Recall
No Early Readmission	0.91	0.28
Readmitted	0.09	0.71

SVM

optimal parameter

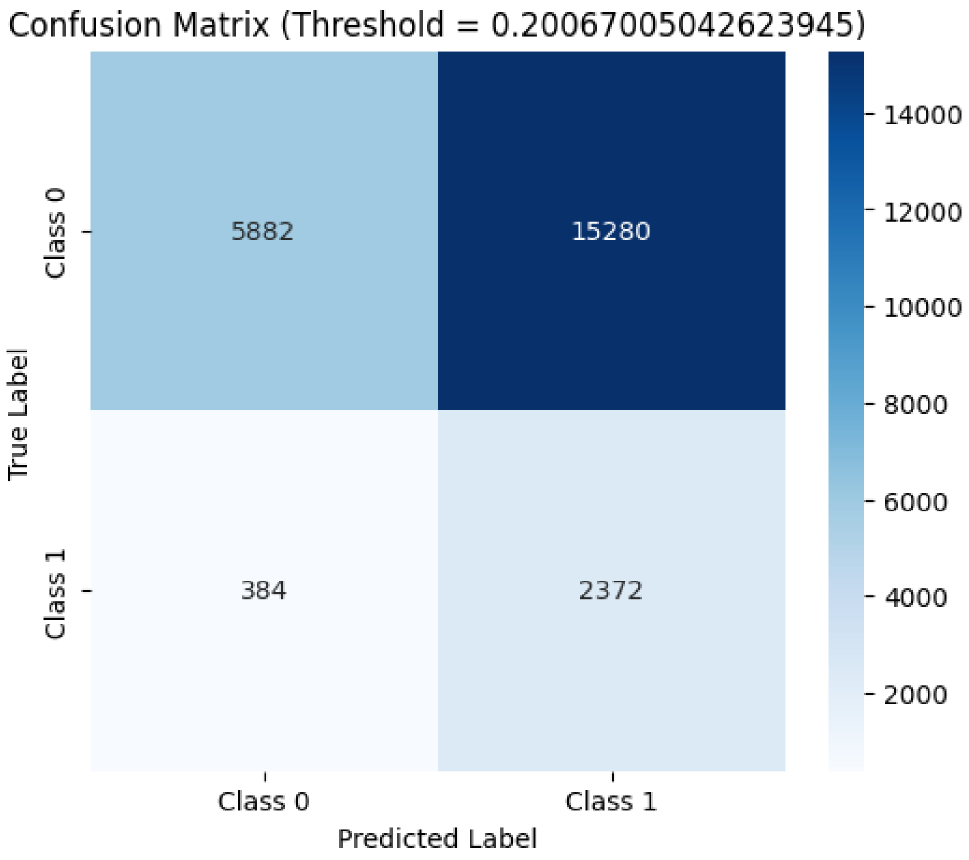
C=0.1, gamma=1, kernel='rb
best decision boundary: 0.11



Storyboard

1) Class imbalance toolkit
    a) SMOTE vs Class Weights
2) Model overview
    a) Gradient Boosting vs Random Forest
    b) XGBoost vs Catboost vs LightGBM
    c) LLM & SVM
3) Hyperparameter optimization
4) Model calibration (clinical use, needs real probabilities)
5) Threshold tuning per F2
6) Threshold tuning against our cost function
    a) financial assumptions
7) Review findings
    a) Process of tuning to F2 vs financial outcomes
8) Discuss implementation
9) Ethical considerations

Financial Impact - Andrew



Class weighting **preserves data distribution**, while **resampling distorts it**, leading to unreliable predictions

Issues with Resampling Methods:

Oversampling:

- Duplicates minority class samples → **Leads to overfitting**
- Model learns an **inflated** positive class proportion → **Overconfident predictions**
- No real improvement in model **generalization**

Undersampling:

- Reduces majority class → **Loss of valuable information**
- Model trained on less data → **Weaker predictive power**
- Results in **high variance & unreliable predictions**

Class Weights: A better alternative

What are Class Weights?

- Instead of **changing the dataset**, we adjust the model's focus by **scaling the loss function**
- Assigns **higher penalty** for misclassifying the minority class
- Model **learns the true distribution** without artificially changing class proportions

Applied in XGBoost Loss Function:

For **binary classification**, the weighted logistic loss function is:

$$\mathcal{L} = -\sum_{i=1}^{N} w_i \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]$$

How it Helps?
Model **does not overestimate** positive class probability
Keeps the **true class distribution intact**
More **balanced confidence estimates**

Attempts to Reduce Dimensionality on Numerical Features

In the aim of reducing dimensionality, **principal component analysis yields minimal results** - most of the **dimensionality should be reduced on the categorical variables**

Feature Correlation Analysis:

Aim: Check for multicollinearity among selected features.

- Most features were uncorrelated, except: **number of medications & time in hospital showed weak positive correlation**
- Decision: No need to drop addition features

Principal Component Analysis:

Aim: Reduce feature space while preserving maximum variance

Results:

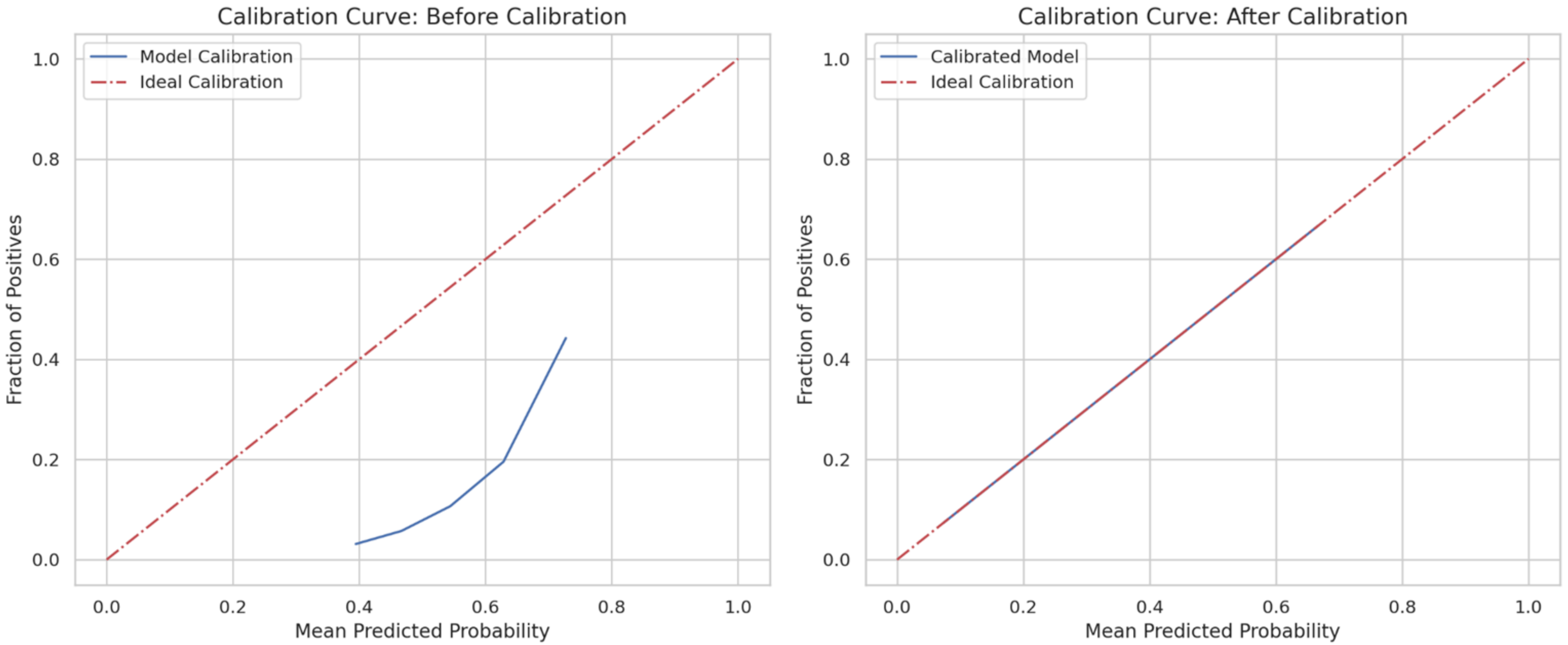
- 3 PCA components explain ~50% variance
- 4 PCA components explain ~66% variance

One-hot encoding on categorical features would introduce upwards of 2,000 features

Model Selection - Hyperparameter Tuned Results

Models	F2-Score	Recall (1)	Precision (1)	Recall (0)	Precision (0)
Random Forest (Optimal Threshold: ~0.20)	0.414	0.861	0.134	0.278	0.939
Logistic Regression					
SVM with class weight (Optimal Threshold: ~0.15)	0.135	0.131	0.148	0.894	0.906
LLM Model(Ollama)	0.294	0.716	0.091	0.283	0.915
XGBoost (Optimal Threshold: ~0.505)	0.426	0.770	0.150	0.450	0.940

Result of Recalibration



Prior to calibration the model was overconfidence, but as the mean probability increases the prediction align more with the actual proportions. In the calibrated model, but the actual prediction and the proportion are aligned.

Threshold tuning helps improve model performance in imbalanced datasets by optimizing F2 score

The default threshold of 0.5 may not be optimal in the case of imbalanced data, as the model tends to predict the negative class more frequently. Therefore, we need to tune the threshold to improve performance on the chosen metric for our specific problem. In this case, we tune the threshold to maximize the F2 score.

Metric	Best Threshold for F2	F1 Score (Class 1)	F2 Score (Class 1)	Precision (Class 0)	Recall (Class 0)	Precision (Class 1)	Recall (Class 1)
Train	0.505	0.2462	0.4266	0.94	0.36	0.14	0.83
Test	—	0.2556	0.4259	0.94	0.45	0.15	0.77

Choosing the Right Metric for Imbalanced Data

Problems with Conventional Metrics

Accuracy is misleading

- Predicting all as "**No Admission**" → **88%** Accuracy
- But **misses all high-risk patients**

F1 Score treats all errors equally

- We need to prioritize recall

Type 2 Error is More Costly !

- Type 1(False positives) → Extra monitoring
- Type 2(False negatives) → Missed Risk
- Severe complications if misclassified

Small icon or graphic showing "False Negative" leading to **hospitalization** vs. "False Positive" leading to **extra checkup**

Better Alternatives: Recall & F2 Score

Why Recall & F2 ?

- Recall: Measures how many actual positives were correctly identified
- F2 score : Prioritizes recall over precision

Formula Comparison

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F2 = \frac{5 \times \text{Precision} \times \text{Recall}}{4 \times \text{Precision} + \text{Recall}}$$

Building a Robust Model for Readmission Prediction

In case if additional rows needed feel free to add models here:

Model	Baseline Modeled Results (F2)	Purpose	Strengths

Building a Robust Model for Readmission Prediction

Predictions will be based on an ensemble model combining outputs from multiple models, **learning the optimal way to combine and give accurate predictions.**

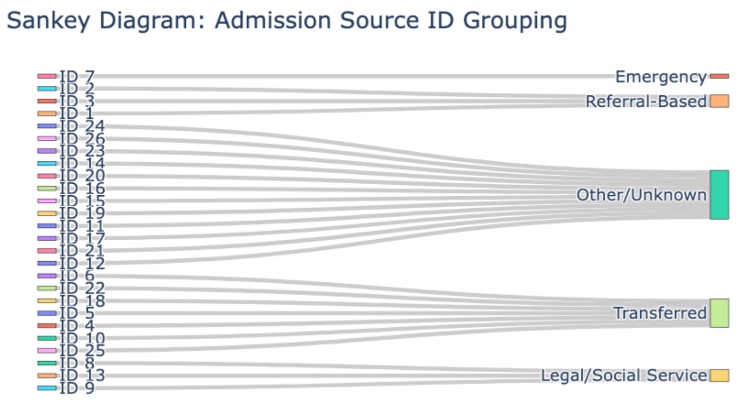
Model	Baseline Modeled Results (F2)	Purpose	Strengths
Random Forest	0.121	Uses bagging to reduce variance	Effective for class imbalance, improves generalization
ExtraTreesClassifier	0.115	Combines multiple decision trees with high randomness	Robust to overfitting, handles complex patterns in imbalanced data
Gradient Boosting (XGBoost, LightGBM, AdaBoost)		Corrects errors iteratively	Captures complex relationships, improves minority class performance
SVM with Class Weights	0.134	Finds optimal decision boundaries	Works well in high-dimensional spaces, focuses on minority class
Logistic Regression with SMOTE		Baseline model with synthetic data balancing	Simple, interpretable, improves dataset balance

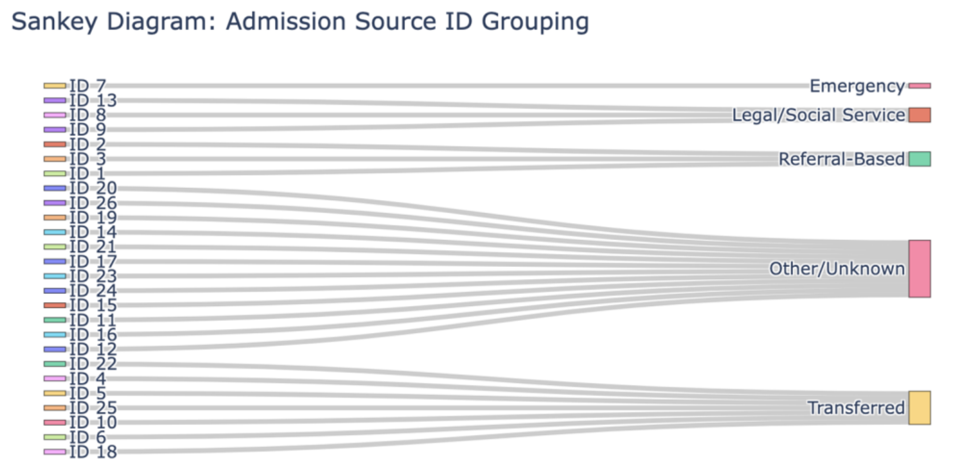
Data Processing - Reducing Categorical Features’ Dimensionality

Contents: Tree Visual for demonstrating how things were grouped and one sentence with short bullets for explaining

For reducing complexity, reducing dimension of the **admission_source_id** from 25 sources to 5 sources

- Combine physician, clinic, HMO referral into a single category (e.g., "Referral-Based").
- Combine transfers from hospital, SNF, healthcare facilities and anything transfer related into a single category (e.g., "Transferred").
- Keep emergency room admission as its own category since it hold distinct importance, (e.g., “Emergency”).
- Combine court/law enforcement, social services into a single category (e.g., "Legal/Social Service").
- Combine other miscellaneous or unknown sources into a single category (e.g., "Other/Unknown").





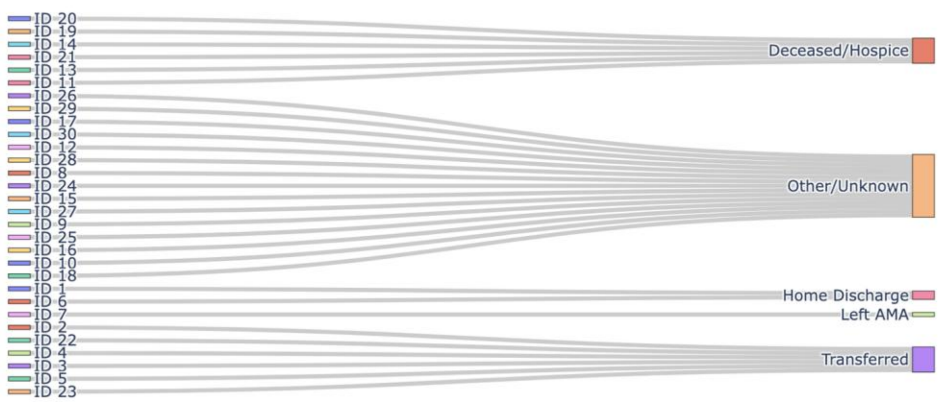
Data Processing - Reducing Categorical Features' Dimensionality

Contents: Tree Visual for demonstrating how things were grouped and one sentence with short bullets for explaining

For reducing complexity, reducing dimension of the **discharge_disposition_id** from 30 dispositions to 5 dispositions

- Combine home and home with health service into a single category (e.g., "Home Discharge").
- Combine transfers to other healthcare facilities into a single category (e.g., "Transferred").
- Keep left against medical advice (AMA) as its own category since it hold distinct importance, (e.g., “Left AMA”).
- Combine deceased or hospice care into a single category (e.g., "Deceased/Hospice").
    - Data points with discharge disposition of “**Deceased/Hospice**” were removed in later cleaning since the **objective is to predict readmissions**, those patients who **died** during this hospital admission were **excluded**. Patients who **passed away** or **hospiced cannot be readmitted**.
- Combine other miscellaneous or unknown cases into a single category (e.g., "Other/Unknown").

Sankey Diagram: Discharge Disposition ID Grouping



Data Processing - Decoding & Grouping Diagnosis By Disease Categories for Better Interpretability

Group Name	ICD-9 Codes	Descriptions
Circulatory	390-459,785	Diseases of the circulatory system
Respiratory	460-519,786	Diseases of the respiratory system
Digestive	520-579,787	Diseases of the digestive system
Diabetes	250.xx	Diabetes mellitus
Injury	800-999	Injury and poisoning
Musculoskeletal	710-739	Diseases of the musculoskeletal system and connective tissue
Genitourinary	580-629,788	Diseases of the genitourinary system
Neoplasms	140-239	Neoplasms
	780,781,784,790-799	Other symptoms, signs, and ill-defined conditions
	240-279, excluding 250	Endocrine, nutritional, and metabolic diseases and immunity disorders, without diabetes
	680-709,782	Diseases of the skin and subcutaneous tissue
	001-139	Infectious and parasitic diseases
Other	290-319	Mental disorders
	E-V	External causes of injury an supplemental classification
	280-289	Diseases of the blood and blood-forming organs
	320-359	Disease of the nervous system

Which Medications Are Important in Prediction?

Problem

- 20+ medications included in the data
- Patients can have extensive history of medications prescribed

Solution

- Reduce the medication features to the list that help predict early readmission

Approach

- Statistical tests for class discrimination (**Chi-square**)
- Filter out medications with extreme class imbalance (e.g., if 95% of patients fall in one category)

Medications Selected:
Metformin
Repaglinide
Glipizide
Insulin

Additional Feature Selection – 2 Layer Robust Approach

Problem

- **40 features** after data processing
- High dimensionality can lead to overfitting and decreased model performance.
- Need for a systematic approach to retain the most informative features.

2 Layer Robust Approach

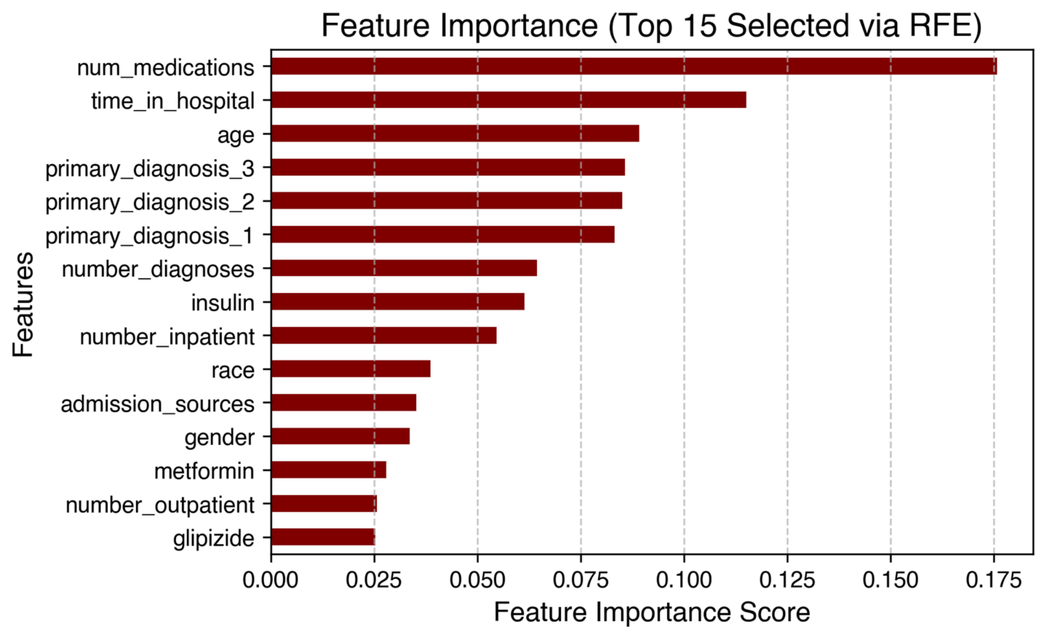
Step 1: Filter Method - SelectKBest with Mutual Information

- Identifies top features using mutual information, which measures the relevance of each feature to the target.
- Selects the **top 30** features based on their **MI scores**.

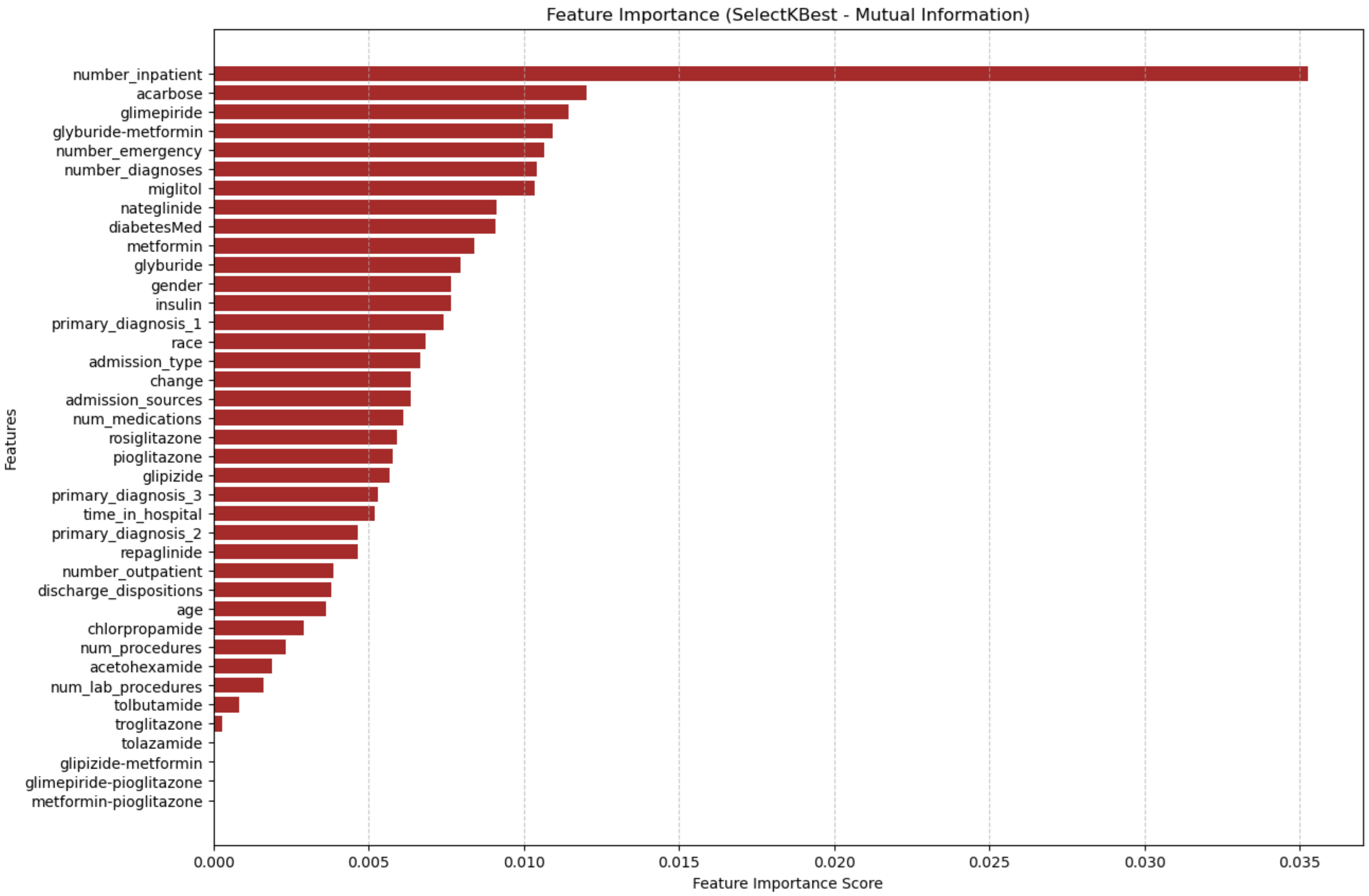
Step 2: Wrapper Method - Recursive Feature Elimination (RFE) with Random Forest

- Uses Random Forest to iteratively remove the least important features.
- Reduces features from 30 to **top 15** most relevant features.

Selected Features & Feature Importance By RFE



Feature Rankings (1st Order Feature Selection Through Filtering)



Feature Rankings (2nd Order Feature Selection Through Wrapping) - Random Forest Ensemble